

Advanced Econometrics

Stationarity and Nonstationarity

Ewa Weychert, PhD

Why stationarity matters

Stationary series

- Constant mean and variance
- Shocks **fade away** over time
- Statistical inference is valid

Non-stationary series

- Mean and/or variance drift
- Shocks **accumulate permanently**
- Regressions can be **spurious**

Key consequence

Before modelling any time series, we must **test** for stationarity.

The unit root – what we are testing for

A series y_t follows an AR(1) process (AR stands for AutoRegressive. The "1" means it uses 1 lag — only yesterday's value to predict today's):

$$y_t = \phi y_{t-1} + \varepsilon_t$$

Condition	Behaviour	Verdict
$ \phi < 1$	Shocks decay	Stationary
$\phi = 1$	Shocks accumulate	Unit root (non-stationary)
$ \phi > 1$	Explosive	Explosive process

The Dickey–Fuller test checks whether $\phi = 1$.

The Role of ϕ (Phi): The Memory Coefficient

Core Concept: ϕ determines how strongly the past influences the present.

The Formula:

$$Y_t = \phi Y_{t-1} + \epsilon_t$$

- **Scenario 1: $\phi = 0.8$ (Mean Reverting)**

- $0.8 \times 100 + \text{noise} \approx 80$
- *Effect:* The series is pulled back toward zero/mean.

- **Scenario 2: $\phi = 1.0$ (Random Walk)**

- $1.0 \times 100 + \text{noise} \approx 100$
- *Effect:* The series has "perfect memory" and wanders aimlessly.

- **Scenario 3: $\phi = 1.1$ (Divergent)**

- $1.1 \times 100 + \text{noise} \approx 110$
- *Effect:* The series "explodes" and grows exponentially.

The Simple Dickey-Fuller (DF) Test

We subtract y_{t-1} from both sides and define $\rho = \phi - 1$:

$$\Delta y_t = \rho y_{t-1} + \varepsilon_t$$

Hypotheses:

- $H_0 : \rho = 0$ (Unit Root / Non-stationary)
- $H_1 : \rho < 0$ (Stationary)

The Flaw: Standard DF assumes ε_t is **White Noise**. If residuals are autocorrelated (errors correlate with past errors), the test statistic τ becomes biased and untrustworthy.

Problem with the standard DF test

Critical assumption

The DF test requires ε_t to be **white noise** – serially uncorrelated.

In practice, real-world residuals are **autocorrelated**.

If autocorrelation is present:

- The τ test statistic is **biased**
- The test will incorrectly classify series too often
- Critical values no longer apply

Solution: Augment the regression with lagged differences.

What autocorrelated errors look like

If ε_t is autocorrelated, it means:

$$\varepsilon_t = \phi_1 \varepsilon_{t-1} + \phi_2 \varepsilon_{t-2} + \cdots + u_t \quad u_t \sim \text{WN}(0, \sigma^2)$$

In other words: **today's error is partly explained by yesterday's error.**

Consequence for DF:

- $\widehat{SE}(\hat{\rho})$ is biased downward
- $|\tau|$ is inflated
- We reject H_0 too often
- Results are **spurious**

Detected by:

Breusch–Godfrey (BG) test on residuals.

If BG p -value $< 0.05 \Rightarrow$
autocorrelation present $\Rightarrow$ do **not** trust τ .

the fix: add lagged differences

We **augment** the DF equation by adding k lagged differences of y :

ADF equation (no constant)

$$\Delta y_t = \rho y_{t-1} + \underbrace{\gamma_1 \Delta y_{t-1} + \gamma_2 \Delta y_{t-2} + \cdots + \gamma_k \Delta y_{t-k}}_{\text{these soak up the autocorrelation}} + \varepsilon_t$$

What changed:

- Added $\sum_{i=1}^k \gamma_i \Delta y_{t-i}$
- Nothing else changed

What did NOT change:

- We still test $H_0 : \rho = 0$
- Same τ critical values
- Same interpretation

What does the *order* mean?

The **order** s tells you *how far back* the correlation reaches.

Order	Condition	Plain meaning
1	$\text{Corr}(\varepsilon_t, \varepsilon_{t-1}) \neq 0$	Today's error correlates with yesterday's error
2	$\text{Corr}(\varepsilon_t, \varepsilon_{t-2}) \neq 0$	Today's error correlates with 2 periods ago
3	$\text{Corr}(\varepsilon_t, \varepsilon_{t-3}) \neq 0$	Today's error correlates with 3 periods ago
p	$\text{Corr}(\varepsilon_t, \varepsilon_{t-p}) \neq 0$	Today's error correlates with p periods ago

Note: order-2 autocorrelation can exist even if order-1 does not – always test multiple orders.

The Breusch–Godfrey test and orders

The BG test checks autocorrelation at a **specific order** p :

$$H_0 : \phi_1 = \phi_2 = \dots = \phi_p = 0 \quad (\text{no autocorrelation up to order } p)$$

We test multiple orders separately because a problem can hide at higher lags:

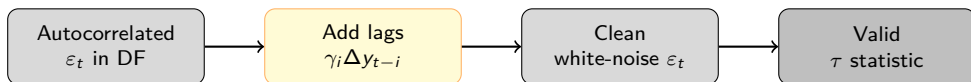
BG order	p -value	Interpretation
1	0.84	No order-1 autocorrelation ✓
2	0.02	Order-2 autocorrelation present – add another lag!
3	0.91	No order-3 autocorrelation ✓

Key point

Passing order 1 does **not** mean you pass order 2. Always check all orders up to $p = 5$ (daily data) or $p = 12$ (monthly data).

Why it works

The lagged differences $\Delta y_{t-1}, \Delta y_{t-2}, \dots$ capture the **dynamic structure** left in the residuals.



Analogy

The lags are **janitors**. They are not being tested.

They just clean the residuals so the test on ρ is trustworthy.

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What is the BG test?

The Breusch–Godfrey test checks whether the **residuals** of a regression are autocorrelated – i.e. whether errors at time t are correlated with errors at earlier times.

Null hypothesis H_0

No autocorrelation in residuals up to order p :

$$\varepsilon_t \perp \varepsilon_{t-1}, \varepsilon_{t-2}, \dots, \varepsilon_{t-p}$$

Residuals are white noise.

Alternative hypothesis H_1

Autocorrelation of *at least* one order exists:

$$\varepsilon_t = \phi_1 \varepsilon_{t-1} + \dots + \phi_p \varepsilon_{t-p} + u_t$$

Residuals are **not** white noise.

Decision rule: if $p\text{-value} < 0.05 \Rightarrow$ reject $H_0 \Rightarrow$ autocorrelation is present.

How the BG test works – the auxiliary regression

Step 1. Estimate the original model, save residuals $\hat{\varepsilon}_t$.

Step 2. Run the *auxiliary regression*:

$$\hat{\varepsilon}_t = \beta_0 + \beta_1 x_{1t} + \cdots + \beta_k x_{kt} + \underbrace{\phi_1 \hat{\varepsilon}_{t-1} + \phi_2 \hat{\varepsilon}_{t-2} + \cdots + \phi_p \hat{\varepsilon}_{t-p}}_{\text{lagged residuals}} + u_t$$

Step 3. Compute the test statistic:

$$\text{LM} = (n - p) \cdot R_{\text{aux}}^2 \xrightarrow{d} \chi^2(p)$$

where n = number of observations, p = lag order tested, R_{aux}^2 = from the auxiliary regression.

Step 4. Reject H_0 if $\text{LM} > \chi_{\alpha}^2(p)$, equivalently if $p\text{-value} < \alpha$.

Choosing the order p to test

What order p means:

Testing whether ε_t is correlated with any of $\varepsilon_{t-1}, \dots, \varepsilon_{t-p}$.

In practice:

- Test several orders: $p = 1, 2, 3, \dots$
- Look at p -values for each
- **All** must be > 0.05 before trusting τ
- Daily data: test up to order 5
- Monthly data: test up to order 12

SP500 lab example – $k = 2$
augmentations

BG order	p -value
1	0.985 ✓
2	0.997 ✓
3	0.980 ✓
4	0.850 ✓
5	0.274 ✓

All $p > 0.05 \Rightarrow$ residuals are white noise
 $\Rightarrow \tau$ is trustworthy.

Summary

Concept	Meaning
Autocorrelation	Error at t is correlated with a past error
Order s	How many periods back the correlation reaches
AR(p) errors	Errors depend on their own p lags
BG at order p	Tests H_0 : no autocorrelation at <i>any</i> lag up to p
Passing all orders	Residuals are white noise $\Rightarrow \tau$ is valid
Failing any order	Add one more lag to ADF and re-test

Rule of thumb

Test BG at orders 1 through 5 for daily data.

Every single one must have p -value > 0.05 before you trust the τ -statistic.

How many lags k to add?

The tradeoff:

Too few lags	Autocorrelation remains $\Rightarrow$ biased τ
Too many lags	Lose observations $\Rightarrow$ loss of power

Goal: smallest k such that residuals are white noise.

SP500 lab result

$k = 0$: BG $p = 0.000035$ (autocorrelation!)

$k = 2$: BG $p > 0.97$ (clean)

How to choose k :

- 1 Start with $k = 0$
- 2 Run BG test on residuals
- 3 If BG $p < 0.05$: add one lag
- 4 Repeat until all BG $p > 0.05$

Or use **AIC** / **BIC** to select k automatically.

Summary – what we do and why

Term	What it is	Why it is there
ρy_{t-1}	Coefficient on lagged level	The thing we test: $H_0 : \rho = 0$
α_1	Intercept	Allows non-zero mean
$\alpha_2 t$	Time trend	Allows deterministic trend
$\sum \gamma_i \Delta y_{t-i}$	Lagged differences	Clean the residuals
ε_t	Residual	Must be white noise

One-line summary

The lagged differences do one job only: they make ε_t white noise so the τ -test on ρ is valid.

Why lags work – the intuition

What the lags do:

- Each $\gamma_i \Delta y_{t-i}$ captures part of the autocorrelation structure
- Once all serial correlation is absorbed, ε_t is white noise
- The τ statistic is now valid

Analogy:

The DF test is a measuring instrument.
Autocorrelation *breaks* the instrument.
Lags *repair* it before we take the reading.

Choosing k

- Too few lags $\rightarrow$ autocorrelation remains $\rightarrow$ biased test
- Too many lags $\rightarrow$ loss of power (harder to reject H_0)
- Check Breusch–Godfrey p -values

Decision rule and critical values

Hypothesis:

$$H_0 : \rho = 0 \quad (\text{unit root})$$

$$H_1 : \rho < 0 \quad (\text{stationary})$$

Reject H_0 if:

$$\tau\text{-statistic} < \text{critical value}$$

(one-sided, left tail)

Typical critical values (Form 2):

	1%	5%	10%
τ_2	-3.43	-2.86	-2.57

Note: DF/ADF critical values differ from the standard t -distribution.

Important

Always verify **no autocorrelation in residuals** (Breusch–Godfrey test) before interpreting the τ statistic.

The Augmented Dickey-Fuller (ADF) Test

To fix autocorrelation, we **augment** the equation with k lagged differences of the dependent variable.

The ADF Equation (with Constant)

$$\Delta y_t = \alpha_1 + \rho y_{t-1} + \underbrace{\sum_{i=1}^k \gamma_i \Delta y_{t-i}}_{\text{The "Janitors"}} + \varepsilon_t$$

Why this works:

- The γ terms "soak up" the dynamic correlation in the data.
- They "clean" the residuals until ε_t is white noise.
- **Result:** The τ -test on ρ becomes valid again.

Practical Procedure for ADF

- ➊ **Start Simple:** Run DF ($k = 0$).
- ➋ **Test Residuals:** Use Breusch-Godfrey (BG) test.
- ➌ **Check p -values:** If BG $p < 0.05$, autocorrelation is present.
- ➍ **Augment:** Add a lag ($k + 1$) and repeat.
- ➎ **Stop:** When all BG tests (up to order 5 for daily data) have $p > 0.05$.

S&P 500 Lab Result

$k = 0 \rightarrow$ BG $p = 0.00003$ (Broken instrument)

$k = 2 \rightarrow$ BG $p = 0.98$ (Clean instrument) $\rightarrow$ **Trust the τ !**

Testing procedure – step by step

- ➊ **Visual inspection** – does the series trend, drift, or appear stationary?
- ➋ **Choose the form** – no constant / constant / constant + trend
- ➌ **Run ADF with 0 augmentations**
- ➍ **Check Breusch–Godfrey** – are residuals autocorrelated?
- ➎ **Add augmentations** until residuals are white noise
- ➏ **Read the τ statistic** and compare to critical values
- ➐ **If H_0 not rejected** – difference the series and repeat from step 1
- ➑ **Determine the order of integration $I(d)$**

Common mistake

Interpreting the τ statistic without first verifying that residuals are free of autocorrelation.

Summary

Test	Null H_0
DF / ADF	Unit root exists
KPSS	Series is stationary
Phillips–Perron	Unit root exists

Key takeaways:

- DF assumes white-noise errors
- ADF adds lags to fix autocorrelation
- Always check BG test before reading τ
- Choose form based on economic reasoning
- Non-stationary $\rightarrow$ difference the series

Order of integration

If level is $I(1)$: first difference is $I(0)$ and safe to model.

Why do we need more than ADF?

The ADF test works well, but has known weaknesses:

ADF limitations

- Sensitive to lag selection
- Low power in small samples
- Null is non-stationarity – we can only *fail to reject*
- Cannot confirm stationarity directly

The solution

Use **multiple tests** with different approaches:

- **PP test** – same null as ADF, non-parametric correction
- **KPSS test** – *reversed* null, directly tests stationarity

If all tests agree $\Rightarrow$ high confidence in conclusion.

Phillips–Perron (PP) test – the idea

The PP test asks the **same question** as ADF:

H_0 : series has a unit root (non-stationary) H_1 : series is stationary

But it handles autocorrelated errors **differently**:

ADF approach:

Add lagged differences to the regression to *parametrically* absorb serial correlation.

$$+ \sum_{i=1}^k \gamma_i \Delta y_{t-i}$$

PP approach:

Keep the simple DF regression, but *non-parametrically correct* the test statistic using a heteroskedasticity and autocorrelation consistent (HAC) variance estimator.

Key advantage

No need to choose lag length k . The correction is automatic.

PP test – the equation and statistic

The PP test is based on the simple DF regression:

$$\Delta y_t = \alpha_1 + \rho y_{t-1} + \varepsilon_t$$

The raw OLS $\hat{\rho}$ is the same – but the test statistic is **corrected**:

$$Z_\tau = \tau \cdot \sqrt{\frac{\hat{\sigma}^2}{\hat{\lambda}^2}} - \frac{(\hat{\lambda}^2 - \hat{\sigma}^2)}{2 \hat{\lambda} \widehat{SE}(\hat{\rho})}$$

τ = raw DF t -statistic

$\hat{\sigma}^2$ = residual variance

$\hat{\lambda}^2$ = long-run variance (HAC estimator, accounts for autocorrelation)

Decision rule – same as ADF

Reject H_0 if $Z_\tau < \text{critical value}$ (critical region: $(-\infty, -2.86)$ at 5%)

PP test in R – SP500 example

```
# PP test on SP500 level
pp.test <- ur.pp(SP500$SP500,
                 type = c("Z-tau"),
                 model = c("constant"))
summary(pp.test)
# Z-tau = 0.363 > -2.86 -> cannot reject H0
# SP500 level is NON-STATIONARY

# PP test on first differences
pp.test.d <- ur.pp(diff.xts(SP500$SP500),
                   type = c("Z-tau"),
                   model = c("constant"))
summary(pp.test.d)
# Z-tau = -70.79 < -2.86 -> reject H0
# diff(SP500) is STATIONARY -> SP500 ~ I(1)
```

KPSS test – the reversed null

The KPSS test is fundamentally different: it **flips the hypotheses**.

ADF / PP

H_0 : **non-stationary** (unit root)

H_1 : stationary

We try to *reject* non-stationarity.
Failing to reject $\neq$ confirming stationarity.

KPSS

H_0 : **stationary**

H_1 : non-stationary (unit root)

We try to *reject* stationarity.
Failing to reject = evidence *for* stationarity.

Critical region – opposite direction!

Reject H_0 when test statistic is **large** (critical region: $(0.463, +\infty)$ at 5%)

KPSS test – how it works

The series is decomposed as:

$$y_t = \alpha_1 + \beta t + r_t + \varepsilon_t$$

where r_t is a **random walk**: $r_t = r_{t-1} + u_t$.

- If $\text{Var}(u_t) = 0 \Rightarrow r_t$ is constant $\Rightarrow$ series is stationary
- If $\text{Var}(u_t) > 0 \Rightarrow r_t$ wanders $\Rightarrow$ series is **non-stationary**

The KPSS statistic tests $H_0 : \text{Var}(u_t) = 0$:

$$\text{KPSS} = \frac{1}{T^2} \sum_{t=1}^T \frac{S_t^2}{\hat{\lambda}^2} \quad S_t = \sum_{j=1}^t \hat{\varepsilon}_j \quad (\text{partial sum of residuals})$$

Large partial sums mean the residuals trend – evidence against stationarity.

KPSS test in R – SP500 example

```
# KPSS test on SP500 level
kpss.test <- ur.kpss(SP500$SP500,
                     type = c("mu")) # "mu" = constant only
summary(kpss.test)
# Statistic = 21.83 > 0.463 (5% CV)
# Reject H0: stationarity -> SP500 is NON-STATIONARY

# KPSS test on first differences
kpss.test.d <- ur.kpss(diff.xts(SP500$SP500),
                       type = c("mu"))
summary(kpss.test.d)
# Statistic = 0.474 slightly above 0.463 (5% CV)
# but below 1% critical value
# Cannot reject H0 at 1% -> diff(SP500) is STATIONARY
# Conclusion: SP500 ~ I(1)
```

KPSS borderline result – Type I error

The KPSS result on `diff(SP500)` was borderline:

	Test statistic		Critical value
5% level	0.474	0.463	slightly exceeds – reject H_0
1% level	0.474	0.739	below – cannot reject H_0

Interpretation

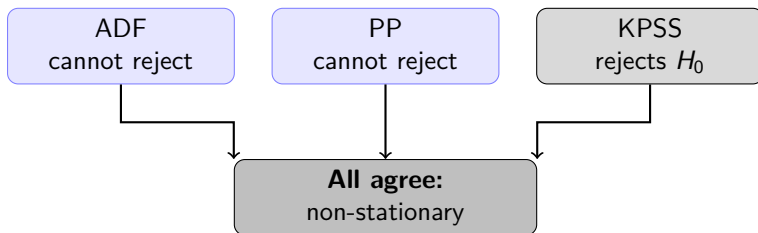
The borderline rejection at 5% is likely a **Type I error** – rejecting a true H_0 .

- Visual inspection of `diff(SP500)` confirms stationarity
- ADF and PP both confirm stationarity of first differences
- We accept the weight of evidence: **SP500** $\sim I(1)$

ADF vs PP vs KPSS – side by side

Property	ADF	PP	KPSS
H_0	Unit root	Unit root	Stationary
H_1	Stationary	Stationary	Unit root
Handles autocorrelation	Parametric (lags)	Non-parametric	Non-parametric
Lag selection needed	Yes	No	No
Reject when statistic	$< CV$ (left)	$< CV$ (left)	$> CV$ (right)
SP500 level	Cannot reject	Cannot reject	Rejects H_0
SP500 diff	Rejects H_0	Rejects H_0	Cannot reject
Conclusion	$I(1)$	$I(1)$	$I(1)$

How to use the three tests together



When tests disagree

If ADF/PP say non-stationary but KPSS says stationary (or vice versa), use visual inspection and economic theory to decide. Prefer the 1% significance level conclusion.

SP500 – complete results across all tests

Series	Test	Statistic	5% CV	Conclusion
SP500 level	ADF	0.274	−1.95	Non-stationary
SP500 level	PP	0.363	−2.86	Non-stationary
SP500 level	KPSS	21.83	0.463	Non-stationary
diff(SP500)	ADF	−50.18	−1.95	Stationary
diff(SP500)	PP	−70.79	−2.86	Stationary
diff(SP500)	KPSS	0.474	0.463	borderline at 5%
Final conclusion				SP500 $\sim I(1)$

Summary

Phillips–Perron (PP):

- Same H_0 as ADF: unit root
- Non-parametric HAC correction
- No lag selection needed
- Reject when $Z_\tau < \text{critical value}$
- Generally more powerful than ADF

KPSS:

- Reversed H_0 : *stationarity*
- Directly tests for stationarity
- Reject when statistic $>$ critical value
- Complements ADF/PP
- Borderline results may reflect Type I error

Best practice

Always run all three tests. If they all agree, the conclusion is robust.

If they conflict, examine the data visually and rely on the 1% significance level.

Thank you

Three forms of the Dickey–Fuller test

Subtract y_{t-1} from both sides, define $\rho = \phi - 1$. Test $H_0 : \rho = 0$ vs $H_1 : \rho < 0$.

Form 1 – No constant, no trend:

$$\Delta y_t = \rho y_{t-1} + \varepsilon_t$$

Form 2 – Constant only (drift):

$$\Delta y_t = \alpha_1 + \rho y_{t-1} + \varepsilon_t$$

Form 3 – Constant + linear trend:

$$\Delta y_t = \alpha_1 + \alpha_2 t + \rho y_{t-1} + \varepsilon_t$$

Choosing the form

Use economic intuition: trending series (GDP) $\rightarrow$ Form 3; mean-reverting series (interest rates) $\rightarrow$ Form 2.